Surprisingly, there is an extremely close relation between Alain Connes's flow for quantum systems and the thermal time that I have discussed above. Connes has shown that, in a quantum system, the thermal flows determined by different macroscopic states are equivalent, up to certain internal symmetries,[86] and that, together, they form precisely the Connes flow.[87] Put more simply: the time determined by macroscopic states and the time determined by quantum noncommutativity are aspects of the same phenomenon.

And it is this thermal and quantum time, I believe,[88] that is the variable that we call "time" in our real universe, where a time variable does not exist at the fundamental level.

The intrinsic quantum indeterminacy of things produces a blurring, like Boltzmann's blurring, which ensures—contrary to what classic physics seemed to indicate—that the unpredictability of the world is maintained even if it were possible to measure everything that is measurable.

Both the sources of blurring—quantum indeterminacy, and the fact that physical systems are composed of zillions of molecules—are at the heart of time. Temporality is profoundly linked to blurring. The blurring is due to the fact that we are ignorant of the microscopic details of the world. The time of physics is, ultimately, the expression of our ignorance of the world. Time is ignorance.

Alain Connes has coauthored with two friends a short science fiction novel. Charlotte, the protagonist, manages to have for a moment a totality of information about the world, without blurring. She manages to "see" the world directly, beyond time:

> I have had the unheard-of good fortune of experiencing a global vision of my being—not of a particular moment, but of my existence "as a whole." I was able to compare its finite nature in space, against which no one protests, with its finite nature in time, which is instead the source of so much outrage.

And then returning to time:

> I had the impression of losing all the infinite information generated by the quantum scene, and this loss was sufficient to drag me irresistibly into the river of time.

The emotion that results from this is an emotion of time:

> This re-emergence of time seemed to me like an intrusion, a source of mental confusion, anguish, fear and alienation.[89]